Case 2 - Conversational Shipment Booking & "Pre-Flight" Error Checker

Booking a cross-border shipment means filling long, confusing forms. Small mistakes in them cause most of the pain later - goods held at customs or units reported missing at the receiving warehouse - and sellers often only find out days later, when their stock is already stuck.

Build a solution that makes booking a shipment easy and catches mistakes before the shipment leaves.

Reference companies - look these up to get a feel for the problem:

- **Flexport** - freight forwarder that lets businesses book and track international shipments online
- **Amazon** - sellers sending stock into Amazon warehouses must label cartons and match counts exactly to what they declared
- **Delhivery** - Indian logistics company offering freight and cross-border shipping for businesses

Key terms

- **Commercial invoice** - the document listing what is in a shipment, how many, and what it is worth; customs relies on it
- **HS code** - an international code that classifies every product for customs
- **Declared value** - the total value of the goods as stated on the paperwork
- **Carton / pallet** - a carton is a box of units; a pallet is a platform holding many cartons
- **Customs hold** - when customs stops a shipment until a problem with it is fixed
- **Delivery appointment** - a booked time slot at the receiving warehouse
- **Transit time** - how many days a shipment takes to arrive

Primer reading: the UK government's guide to creating an export (commercial) invoice (https://www.business.gov.uk/export-from-uk/advice/manage-payment-for-export-orders/how-to-create-an-export-invoice).

Test data format

The template (*Case2_Shipment_Booking_Test_Data_Format.xlsx*) has four sheets. **Sheets 3 and 4 are fixed and already filled in** - the carrier options and the rulebook below are exactly what judges will use, so you can build around them. Judges will upload only Sheets 1 and 2, in exactly this structure. Every test shipment goes from the UK to Germany, France or the Netherlands.

Not sure what a column means? The template has a **Glossary** sheet explaining every column that is not self-explanatory.

Allowed values - test data will only use the values below, so you do not need to handle anything else.

Field	Allowed values
Origin country	United Kingdom

Field	Allowed values
Origin city	London, Manchester, Birmingham, Leeds
Destination country	Germany, France, Netherlands
Destination city	Germany: Berlin, Frankfurt, Hamburg · France: Paris, Lyon · Netherlands: Amsterdam, Rotterdam
Currency	GBP only
Country of origin (items)	United Kingdom, China, India, Vietnam, Turkey, Germany

How to read the tables below: judges fill in every column in Sheets 1 and 2 - your solution only reads them and never has to create them. Sheets 3 and 4 are fixed and already filled in. **Fixed list** columns only use the values shown. **Any text** columns can contain anything the seller wrote. **ID** columns are just reference labels. Values marked "as the seller states it" may be wrong - checking them is your job.

Sheet 1 - Bookings (one row per shipment)

Column	What it means	Format	Example
Booking ID	Reference label for each shipment	ID	BK-001
Seller name	Business sending the goods	Any text	Northfield Goods
Origin city · Origin country	Where the goods are collected from	Fixed list (see Allowed values)	Manchester · United Kingdom
Destination city · Destination country	Where the goods are delivered	Fixed list (see Allowed values)	Frankfurt · Germany
Destination type	Kind of site receiving the goods	Fixed list: Warehouse / Business address	Warehouse
Ready to ship date	Day the goods will be packed and ready for collection	Date, DD-MM-YYYY	05-11-2026
Seller priority	What matters most to the seller. Most reliable = highest on-time rate; Balanced = no strong preference, best trade-off of cost, speed and reliability	Fixed list: Cheapest / Fastest / Most reliable / Balanced	Cheapest
Pallets	Number of pallets, as the seller states it	Number	3

Column	What it means	Format	Example
Cartons	Total cartons, as the seller states it	Number	90
Total declared value	Total value of the goods, as the seller states it	Number	5400
Currency	Currency of every money value	Fixed: GBP	GBP
Total gross weight (kg)	Weight of everything shipped, incl. cartons and pallets, as the seller states it	Number	620
Delivery appointment booked	Whether the seller has a delivery time slot booked at the receiving site	Fixed list: Yes / No	Yes

Sheet 2 - Items (one row per product in a booking)

Column	What it means	Format	Example
Booking ID	Which booking this item belongs to; matches Sheet 1	ID	BK-001
SKU · Description	Seller's product code and a plain description	Any text · Any text	MUG-01 · Ceramic coffee mug, 350 ml
HS code	Customs code for the product, as the seller states it	Number, may be blank	691200
Quantity (units)	Total units of this product, as the seller states it	Number	1080
Unit price	Price of one unit	Number, GBP	5.00
Units per carton	Units packed in one carton	Number	12
Cartons	Cartons of this product, as the seller states it	Number	90
Unit weight (kg)	Weight of one unit	Number	0.5
Country of origin	Where the product was made - not where it ships from	Fixed list (see Allowed values), may be blank	United Kingdom

Sheet 3 - Carrier options (fixed - see the template for the full rate card)

Column	What it means	Format	Example
Carrier · Service	Carrier name and its service level	Fixed (in template)	Carrier A · Road economy
Origin country → Destination country	Route the service runs on	Fixed (in template)	United Kingdom → Germany
Price per pallet	Cost to ship one pallet	Fixed (in template), GBP	180
Transit time (days)	Days from collection to delivery	Fixed (in template)	5
On-time rate (%)	Share of shipments that arrive on the promised day	Fixed (in template)	92
Max weight per pallet (kg)	Heaviest pallet the service accepts	Fixed (in template)	500

Sheet 4 - Rulebook (fixed - these are the only rules we test against)

Rule ID	Rule	Severity if broken
R-01	Every item must have an HS code of at least 6 digits	Red
R-02	Total declared value must equal the sum of quantity × unit price across all items (within 1%)	Red
R-03	For each item, units per carton × cartons must equal quantity	Red
R-04	Total cartons on the booking must equal the sum of cartons across all items	Red
R-05	Weight per pallet (total gross weight ÷ pallets) must not exceed the chosen carrier's max weight per pallet	Red
R-06	No lithium batteries, aerosols or flammable liquids	Red
R-07	Total gross weight must be at least the sum of unit weight × quantity across all items	Amber
R-08	Shipments to a warehouse must have a delivery appointment booked	Amber
R-09	Every item must have a country of origin	Amber
R-10	Ready to ship date must not be in the past	Amber

How we will test your MVP link

Testing happens in two parts.

Part 1 - Conversation (no file)

- We will describe a shipment in plain language, the way a seller would, often leaving details out.
- We expect your solution to ask only for what is missing and end up with a complete booking.
- We will change a detail mid-conversation (e.g., add a pallet) and check that the booking and its checks update.

Part 2 - File upload

- We will upload our own Bookings and Items sheets in the format above. Some bookings will contain planted mistakes that break the rulebook.

For every booking, from either part

- We expect to see what is wrong, how serious it is, and how to fix it.
- We expect a recommended carrier option based on the seller's priority, with the reason, then a confirmation the seller can understand and what happens next.
- We judge your reasoning and how clearly your solution explains its decisions - there is no single right answer.
- The link must work for us without your help or a login.